

RYAN CHIN JIAN HWA

WP Kuala Lumpur, Malaysia

☎ +6010-9732858 ✉ ryanchinh@gmail.com 🔗 [linkedin.com/in/ryanchinh](https://www.linkedin.com/in/ryanchinh) 📄 github.com/Wrynaft

Education

Universiti Malaya

Oct. 2023 – 2027

Bachelor of Computer Science (Data Science) | CGPA: 3.97

- **Honors:** Dean's List (Every Semester)
- **Scholarship:** Jabatan Perkhidmatan Awam (JPA) LSPM

Ryukoku University, Kyoto

Spring Semester 2026

Japan Studies Program

Relevant Experience

Faculty of Computer Science & Information Technology | Universiti Malaya

Teaching Assistant – Data Structures

Feb. 2025 – Present

- Assisted lecturer part-time in conducting lab sessions and explaining data structures concepts using Java.

Teaching Assistant – Fundamentals of Programming

Oct. 2024 – Jan. 2025

- Facilitated lab test and assignment review of students by contributing to setting test questions and giving fair evaluations.

Projects

LLM Multi-Agent System for Stock Portfolio Construction | Python, Gemini LLM, MongoDB

Jan. 2026

- Developed an LLM-based multi-agent system using Gemini 2.0 Flash to automate stock portfolio construction and optimization for the Bursa Malaysia market.
- Architected a 4-agent framework (Coordinator, Fundamental, Valuation, Sentiment) utilizing Chain-of-Thought (CoT) prompting and Round Robin debates to generate explainable, risk-aware investment recommendations.
- Engineered a data pipeline extracting historical prices via yfinance and scraping financial news using BeautifulSoup, structuring over 1.1 million records into MongoDB.
- Achieved 100% tool selection accuracy and outperformed the KLCI benchmark in 1-month back-testing for risk-averse profiles. Deployed the interactive system as a web application via Streamlit.

Analysis of Public Transport Ridership in Klang Valley | Python, PowerBI

Apr. 2025

- Preprocessed data through data integration from multiple sources and feature generation such as moving averages.
- Generated analytics including data visualizations about trends and factors affecting ridership, using Matplotlib and Seaborn.
- Engineered and compared predictive models to forecast future trends of ridership.

Skills & Certifications

Programming Languages: Python, SQL, Java, R, TypeScript, Dart, HTML

AI & Data Science: Large Language Models (Gemini), Multi-Agent Systems, Prompt Engineering, NLP (FinBERT), Pandas, Matplotlib, Scikit-learn

Frameworks & Tools: Streamlit, Angular, Spring Boot, Flutter, BeautifulSoup, Jupyter, PowerBI, Tableau, Apache Spark, Apache Hadoop

Databases: MongoDB (NoSQL), MySQL

Certifications: AWS Academy Cloud Foundations (2024), Google Cloud Data Analytics Certificate (2026), Google Advanced Data Analytics Professional Certificate (2026)

Spoken Languages: Fluent in English, Bahasa Malaysia, and Mandarin

Featured Awards

Consolation Prize (Top 10) & Best Interactive Dashboard | UKM Data Challenge 4.0

May 2025

Consolation Prize | Data Challenge 3.0 (UKM Statistics Club)

May 2024

SPM Top Scorer Award | 11A+, Chung Ling High School

2022

Leadership / Extracurricular

Persatuan Komputer Universiti Malaya (PEKOM)

Oct. 2024 – Jan. 2025

Head of Contest Protocol Department – UM Game Jam 2024

- Developed clear contest rules, a contest theme, and challenges to ensure a fair and smooth competition.
- Managed department members by involving them in active discussions on decision-making, delegating tasks, and facilitating progress.